

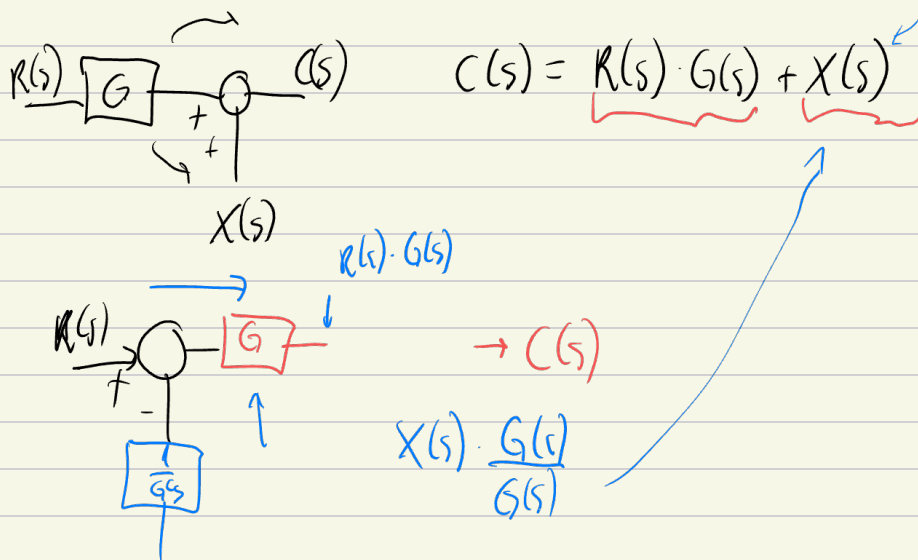
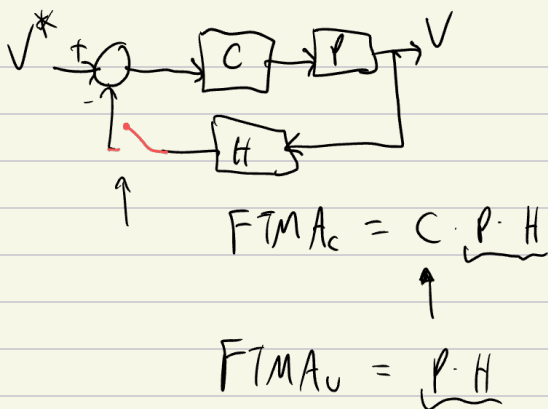
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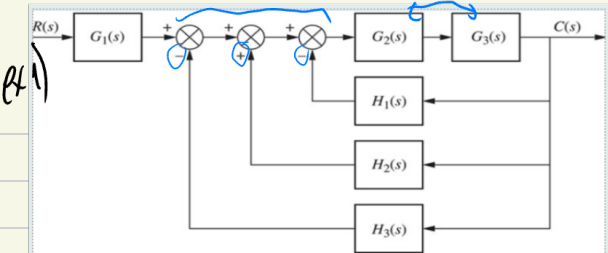
Controle para Sistemas
Eletrônicos de Potência

Prof. Tiago Davi Curi Bosarello

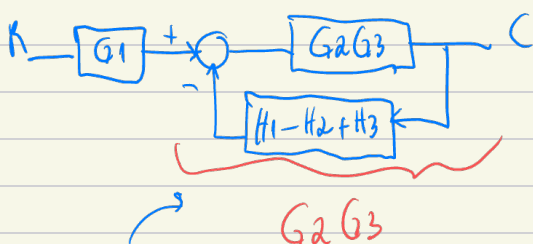
2026-2



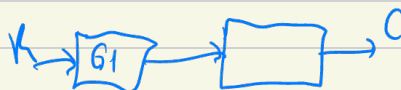




$$\frac{C(s)}{R(s)} = ?$$

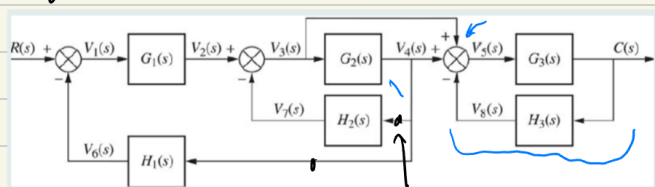


$$\frac{C(s)}{R(s)} =$$

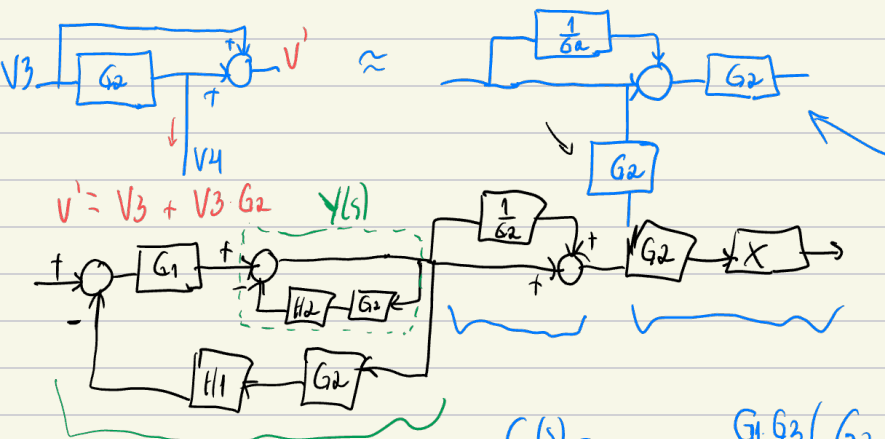


$$\frac{G_2 G_3}{1 + G_2 G_3 (H_1 - H_2 + H_3)} \quad \xrightarrow{\quad} \quad \frac{G_1 G_2 G_3}{1 + G_1 G_3 (H_1 - H_2 + H_3)}$$

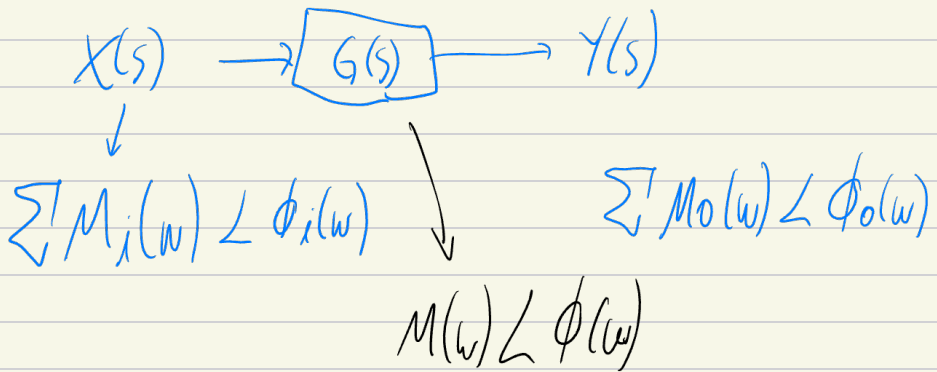
Pl 2



$$X(s) = \frac{G_3}{1 + G_3 H_3}$$

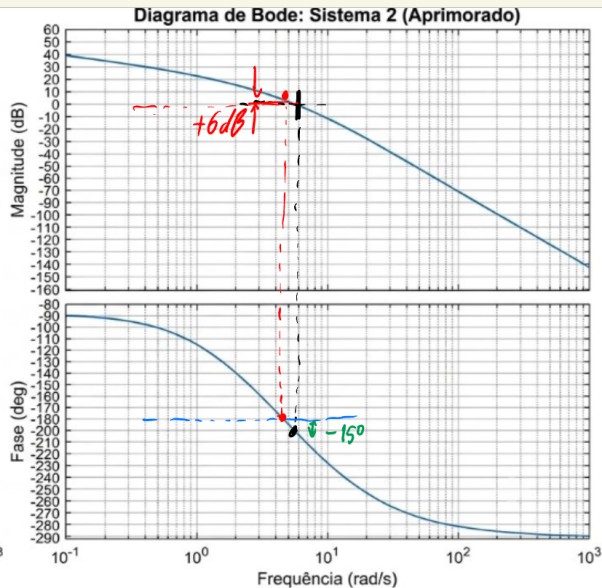
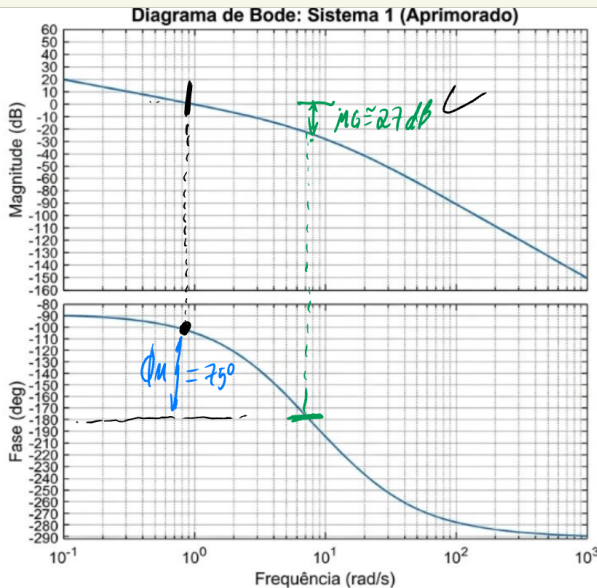


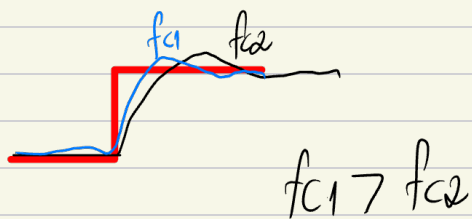
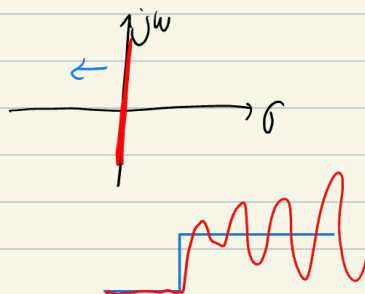
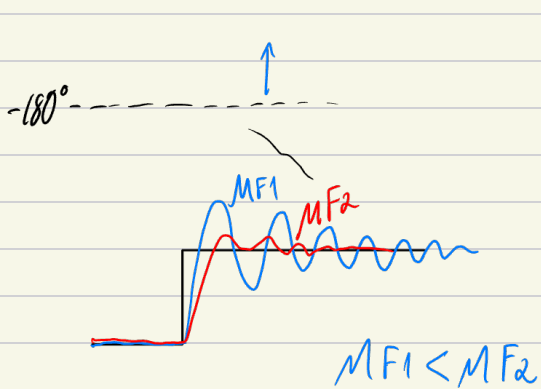
$$\frac{C(s)}{R(s)} = \frac{G_1 G_3 (G_2 + 1)}{\{1 + G_2 H_2 + G_1 G_2 H_1\} \{1 + G_3 H_3\}}$$



$$M_o = M_i \cdot M$$

$$\phi_o = \phi_i + \phi$$





27/8/26

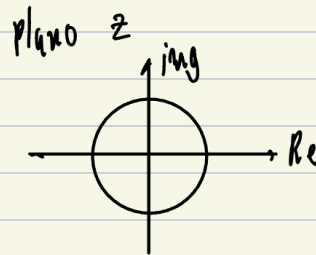
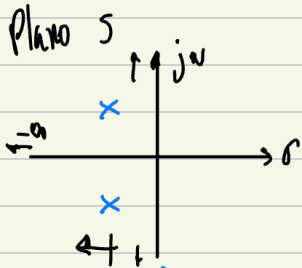
- Diagrama de blocos
- Resposta em freq.
 - estabilidade, erro
 - MF (oscilação da resposta)
 - fc (velocidade)

Modelagem $\rightarrow$ Projetar do controlador $\rightarrow$ Discretização $\rightarrow$ Controle Digital

s s

Discretização de Controladores

$$X(s) \rightarrow X(z)$$



relação entre os planos

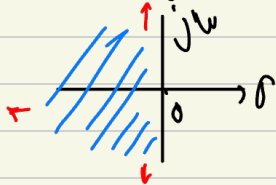
$$z = e^{T \cdot s} \quad \text{onde } T \text{ é o período de amostragem}$$

⇒ Mapeamento do lado esquerdo do plano s no plano z

$$z = e^{Ts}$$

$$s = \sigma + j\omega$$

$$z = e^{T(\sigma + j\omega)} = e^{T \cdot \sigma} \cdot e^{jT\omega} = e^{T \cdot \sigma} \cdot e^{j(T\omega + 2k\pi)}$$



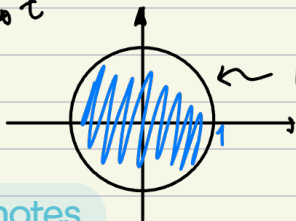
$$\sigma < 0$$



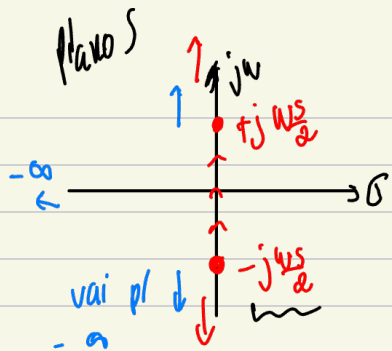
$$|e^{T \cdot \sigma}| < 1 \quad |z| = 1$$

$$\sigma = 0 \quad e^{T \cdot \sigma} = 1 \quad z = 1 \cdot e^{jT\omega} \quad |z| = 1$$

plano z



← limite da estabilidade



$$z = e^{j\omega T}$$

$$\angle \omega \cdot T$$

↑
varia $-\infty$ a $+\infty$
conforme ω varia de $-\infty$
a $+\infty$

$$\omega = 2\pi f$$

$$f = \frac{1}{T}$$

$$T = \frac{2\pi}{\omega_s}$$

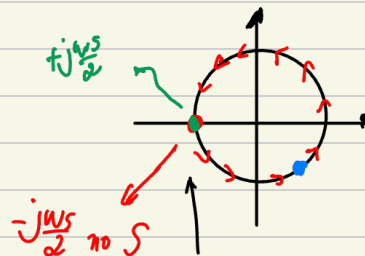
$$z = e^{-j\frac{\omega_s}{2}T} = e^{-j\frac{\omega_s}{2} \cdot \frac{2\pi}{\omega_s}} = e^{-j\pi}$$

$$1 \cdot e^{-j\pi} \quad \left| \begin{array}{l} |z| = 1 \\ \angle z = -\pi \end{array} \right.$$

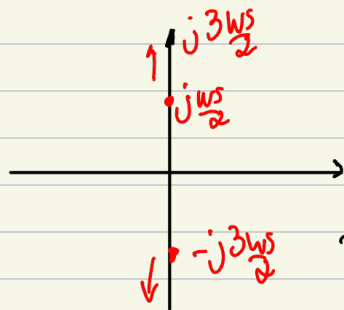
$$p/s = j\frac{\omega_s}{2}$$

$$z = e^{j\frac{\omega_s}{2}T}$$

$$|z| = 1 \quad \angle z = \pi$$



e se continuar indo pl cima



$$s = j3\frac{\omega_s}{2}$$

$$\sim |z| = 1 \\ \angle z = -3\pi$$

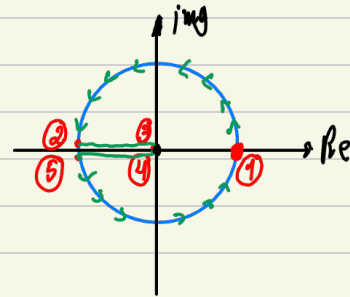
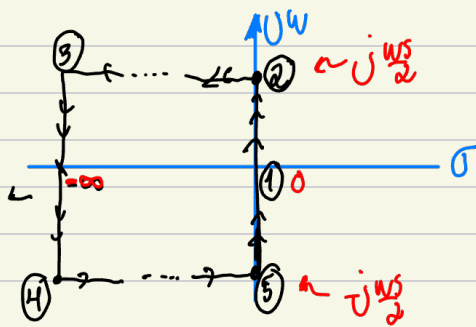
$$z = e^{sT} = e^{+j3\frac{\omega_s}{2} \cdot \frac{2\pi}{\omega_s}} = e^{+j3\pi}$$

$$|z| = 1$$

$$\angle z = +3\pi$$

du 3
volts

plano S.



Ponto 1: $s=0$
 $z = e^{sT} = e^{0 \cdot T} = 1$

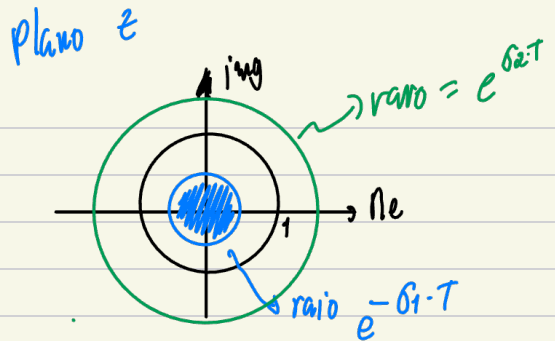
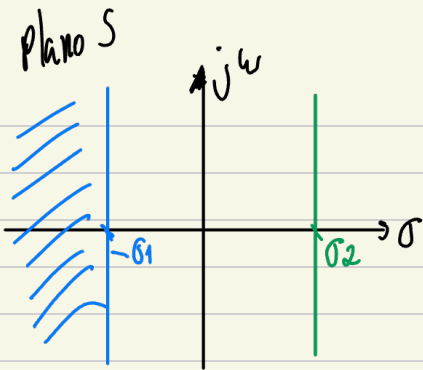
Ponto 2 e 5:
 $\angle z = -\pi$ e π

Ponto 3
 $s = -\infty + j\frac{\omega_s}{2}$

$z = e^{sT}$
 $z = e^{-\infty T} \cdot e^{j\frac{\omega_s}{2} T}$
 $z = 0$

Ponto 4
 $s = -\infty - j\frac{\omega_s}{2}$

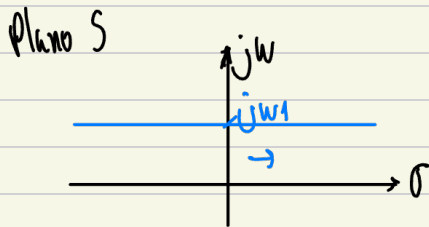
$z = 0$



$$s = -\sigma_1 + j\omega T$$

$$z = e^{sT} = e^{-\sigma_1 \cdot T} \cdot e^{j\omega T}$$

$$|z| = \underline{e^{-\sigma_1 \cdot T}}$$



σ variando de $-\infty$ a $+\infty$



$$z = e^{sT}$$

$$s = \sigma + j\omega T$$

$$z = e^{(\sigma + j\omega T)T} = e^{\sigma T} \cdot \underline{e^{j\omega_1 T}}$$

$$\angle z = \omega_1 \cdot T$$

p/ $\sigma < 0$ $|z| = e^{\sigma T} < 1$ $z = e^{-\infty T} \rightarrow 0$
 z dentro do círculo

p/ $\sigma = 0$ $|z| = 1$

p/ $\sigma > 0$ $|z| = e^{+\infty \cdot T} \rightarrow \infty$

Métodos de Discretização

→ Mover $H(s)$ p/ $H(z)$

→ A discretização consiste em aproximar uma eq. diferencial por uma eq. à diferença.

a) Forward

$$x(t) = \left. \frac{dy(t)}{dt} \right|_{t=kT} \approx \frac{y[(k+1)T] - y(kT)}{T} = x(k)$$

$$x(k) = \frac{y(k+1) - y(k)}{T} \Rightarrow \text{Aplicando a transformada } z$$

$$X(z) = \frac{zY(z) - Y(z)}{T} = Y(z) \frac{z-1}{T}$$

$$\left[\begin{array}{l} \frac{Y(z)}{X(z)} = \frac{1}{\frac{z-1}{T}} \\ \frac{Y(s)}{X(s)} = \frac{1}{s} \end{array} \right]$$

$$x(t) = \frac{dy(t)}{dt} \quad y(t) = \int x(t) \quad Y(s) = \frac{X(s)}{s}$$

$$\frac{1}{s} = \frac{1}{\frac{z-1}{T}}$$

$$\therefore s = \frac{z-1}{T}$$

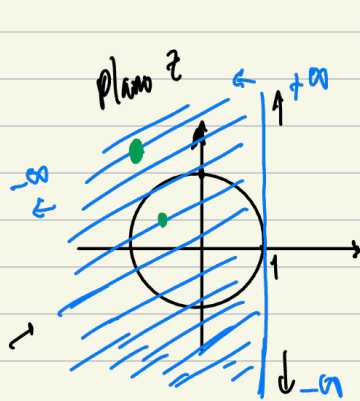
$$s = \frac{z-1}{T}$$

Mapeamento no plano z

isolando o z

$$z = 1 + sT = 1 + j\omega T$$

$\uparrow$ fixa $\uparrow$ variável



$p/ \omega = 0$	$z = 1$
$p/ \omega \rightarrow \infty$	$z = +\infty$
$p/ \omega \rightarrow -\infty$	$z = -\infty$
$p/ \omega = \frac{1}{T}$	$z = 1 + j$
$p/ \omega = \frac{1}{T}$	$z = 1 + j^4$
	$ z < 1$

b) Backward

$$x(k) = \frac{dy(k)}{dt} \approx \frac{y(kT) - y((k-1)T)}{T} = x(k)$$

$$x(k) = \frac{y(k) - y(k-1)}{T} \quad z \quad \Rightarrow \quad X(z) = \frac{Y(z) - z^{-1}Y(z)}{T}$$

$$\frac{Y(z)}{X(z)} = \frac{T}{1 - z^{-1}} = \frac{T \cdot z}{z - 1} = \frac{Y(s)}{X(s)} = \frac{1}{s}$$

$$s = \frac{z-1}{T \cdot z}$$

Mapeando no plano z

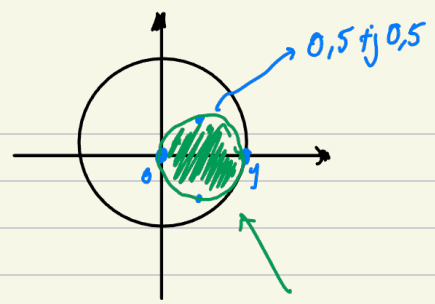
$$s = \frac{z-1}{T \cdot z}$$

$$z = \frac{1}{1-sT} \quad s = j\omega$$

$$z = \frac{1}{1-j\omega T} \cdot \frac{(1+j\omega T)}{(1+j\omega T)} = \frac{1+j\omega T}{1+\omega^2 T^2}$$

$$\text{Parte real} = \frac{1}{1+\omega^2 T^2}$$

$$\text{Parte imag} = \frac{\omega T}{1+\omega^2 T^2}$$



$p/w = 0$	$z = \frac{1}{1-j0} = 1$	$p/w = -\frac{1}{T}$
$p/w \rightarrow \infty$	$z = \frac{1}{1-j\infty} = 0$	$z = 0.5 - j0.5$
$p/w = \frac{1}{T}$	$z = \frac{1}{1-j\frac{1}{T} \cdot T} = \frac{1}{1-j} = 0.5 + j0.5$	